

Statistical Research Consulting
Statistical Consulting Center, Central Michigan University
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This document presents 16 selected research consultancy projects in which I provided statistical consulting to faculty, graduate students, and external collaborators across diverse disciplines, with a focus on study design, data management, and advanced statistical modeling.

Statistical Methods: Hypothesis testing; independent-samples t-test; Wilcoxon rank-sum test; paired t-test; Wilcoxon signed-rank test; chi-square test; Fisher's exact test; exact binomial test; two-proportion z-test; McNemar's test; Pearson correlation; Spearman correlation; one-way ANOVA; two-way ANOVA; mixed-design ANOVA; linear mixed-effects models; linear regression; proportional odds models; logistic regression with interaction effects; Firth logistic regression; generalized linear models (GLMs) with interaction effects; trend tests; intraclass correlation.

Data Management & Analysis Tools: SAS, R, SPSS, and Microsoft Excel.

Domains: Public health and clinical-related research (including clinical studies); geography and environmental sciences; business administration; library and information science.

No.	Project Title	Client	Department	Year	Services	Statistical Methods	Software
1	Medicare Hospital Cost Report Data Extraction and Management	Yuriy Kutny, DHA Cohort 2023	Healthcare Administration	2026	Data extraction; data management; development of data dictionary		SAS
2	Efficacy of omalizumab patients with severe urticaria and concomitant atopic dermatitis	Jenna Kaitlyn Currier (Medical Student)	Medicine	2026	Statistical modeling; interpretation	Binomial test; chi-square; McNemar's test	R
3	PHQ--BIMS--Socialization Study	Rochelle M. Kopka, DHSc	Physical Therapy	2025--2026	Data cleaning; statistical modeling; interpretation	Paired t-test; correlation; ANOVA	SAS
4	Body Image Dissatisfaction and Its relationship with Physical Activity and disordered eating Behaviors among a sample of CMU Students.	Najat Yahia, PhD; Anna Moore (Student)	Rehabilitation and Medical Sciences	2025--2026	Data cleaning; statistical modeling; interpretation; poster contribution	ANOVA; regression; chi-square	SAS
5	Climate Change & Gender Identity Study	Anthony Feig, PhD; Aiden Kosciesza, PhD	Geography and Environmental Studies/Communication	2026	Data cleaning; statistical modeling; interpretation of results	Proportional odds model; Spearman correlation	SAS

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6	The Behavioral Workplace Impact of the Leadership Standards Initiative Training	Amy McGinnis, EdD; Morgan Shropshire (Student)	Business Administration	2025	Statistical consulting; modeling	Chi-square; t-test; regression; correlation	SPSS
7	Experiences of d/Deaf and Hard of Hearing Academics	Anthony Feig, EdD	Geography and Environmental Studies	2025	Advising on statistical modeling; data cleaning; EDA; interpretation of results	Exploratory data analysis (EDA)	SAS
8	DAS21 Survey for Doctor of Physical Therapy Students	Emily Schubbe, EdD	Physical Therapy	2025	Data cleaning; statistical modeling; interpretation	GLMs with interaction effects	SAS
9	Digital Learning Faculty Fellows	Katie Edmiston, MLIS	Library	2025	Study design; survey review; statistical modeling; interpretation of results	t-test; Wilcoxon; chi-square; Fisher's exact test	SAS
10	The Informal Competitor Next Door: The Evolving Nature of Formal Firm Impact & Response in Paraguay	Michael Pisani, PhD	Management	2025	Panel data analysis; statistical modeling; interpretation of results	Logistic regression with interaction effect	R
11	AI vs Medical Student Performance in Problem-Based Tasks in Basic Sciences	Brianne Lewis, PhD; Umme Aiman Mirza (Medical Student)	Medicine	2025	Statistical modeling; manuscript preparation	Intraclass correlation (ICC); linear mixed-effects model	R
12	First-Year Transfer Retention Study between Students of Color and White Students at a Four-Year Predominantly White Institution	Eric Buschlen-Shawntain Jenkins (EdD Student)	Education and Human Services	2024	Statistical consulting; modeling strategy	Pearson correlation; ANOVA	SPSS
13	The Prognostic Nutritional Index in Severe Alcohol Withdrawal Syndrome	Juliette Perzhinsky, MD, MSc, FACP, FASAM; Makenzie Wank (Medical Student)	Medicine	2024	Statistical modeling; interpretation of results	Proportional odds model; logistic regression; Firth logistic regression	SAS
14	Effect of Clinical Span of Control on Population Health Outcomes & Member Experience	Steven Berkshire, DHA, Professor Emeritus; Britany K. Leiendecker (Student)	Health Professions	2024	Study design advising; statistical modeling; interpretation	Hypothesis testing; mixed-design ANOVA	SPSS

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15	Assess Medical Student Knowledge of Physical Therapy Services	Rochelle M. Kopka, DHSc; Keira Stampfly (Student)	Physical Therapy	2024	Statistical analysis; interpretation of results	Paired t-test; Wilcoxon signed-rank test	SAS
16	The Effect of Blinding Faculty Reviewers on Increasing Diversity in Residency Recruitment	Dominique Hill, MD, FACEP	Emergency Medicine, Internal Medicine, Psychiatry	2024	Statistical analysis; interpretation of results	Cochran--Armitage trend test; Cochran--Mantel--Haenszel test	SAS